Alignment, position-based, and hybrid models of locust collective motion: a four-metric calibration against field trajectories

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Abstract

We test which local interaction rule reproduces the collective motion of free-ranging locust hopper bands. We simulate the Vicsek alignment model [7], a position-based pull model after Sayin et al. [6], and a sequence of hybrids that combine forward-weighted alignment, anisotropic noise, heading inertia, and a soft preferred-spacing zone. We compare each model to the trajectory dataset of Weinburd et al. [8] across four metrics: polarization Φ, turning-angle standard deviation, nearest-neighbor distance, and an anisotropy index of the body-centered neighbor density evaluated both globally and inside a near-zone radial band. The two baseline models fail in distinct, mechanistic ways: Vicsek matches polarization only at unrealistic noise levels and produces no spatial structure; the simplified pull model collapses to a disordered steady state from any initial condition, for a geometric reason that we argue rules out an entire class of instantaneous position-based rules. Hand-tuning a hybrid closed some gaps but introduced others. We replace hand-tuning with the Covariance Matrix Adaptation Evolution Strategy (CMA-ES) [4], parallelized across CPU cores, and search a six- and seven-dimensional parameter space against a weighted relative-error loss. The calibrated six-parameter hybrid (v4) matches polarization within 10%, turning-angle std and nearest-neighbor distance within 1%, and reproduces the field's body-centered neighbor density both globally ($A_{\text{global}} \approx 0$) and inside a 1.5–3.0 cm near-zone band ($A_{\text{near}} = -0.005$ for both v4 and the field). A seven-parameter variant adding temporal bearing integration (v5) reaches a marginally lower scalar loss but lands in a basin where the preferred-spacing ring carries a forward-bias artifact ($A_{\text{near}} = -0.06$) that the field does not have; the seventh parameter is unnecessary at the field's measurement scale and worsens the spatial fit. The calibrated fit depends on periodic boundary conditions; under reflective walls it breaks. Because real hopper bands are open systems, we treat the calibrated parameters as a successful fit to a particular simulation regime rather than a demonstration that this rule structure governs field hopper-band dynamics.

1 Introduction

Australian plague locusts (*Chortoicetes terminifera*) form hopper bands that march coherently across kilometers of terrain, causing agricultural damage at continental scale. Two competing theories of how they coordinate have appeared in the literature. The first, originating with Vicsek et al. [7] and applied to locusts by Buhl et al. [2], proposes that each individual aligns its heading with neighbors inside a fixed radius. Order arises from a density-driven phase transition. The second, advocated by Sayin et al. [6] on the basis of field, laboratory, and virtual-reality experiments with desert locusts (*Schistocerca gregaria*), proposes that individuals are attracted to the positions of forward neighbors. The interaction is geometric rather than directional, and Sayin et al. [6] argue that bearing information is integrated over time by a neural ring-attractor circuit, rather than read out instantaneously. We test models inspired by both theories against *C. terminifera* trajectory data; the species mismatch is a caveat worth noting, although the two species share the gregarious-phase marching behavior the theories aim to explain.

These theories make different predictions about the spatial structure of the swarm and about the frame-to-frame heading dynamics of individuals. Until recently they could not be tested against individual-level field data. Weinburd et al. [8] released the first such dataset: roughly 3.3 million position records and 20,000 trajectories from four hopper bands, sampled at 25 Hz. We use this dataset as the ground truth.

We ask which local rule, or which combination of rules, reproduces the field measurements. Our contribution is threefold. First, we show that both pure alignment and pure pull fail in characteristic ways. The pull failure is sharp: starting from perfect alignment ($\Phi = 1$), the model collapses to the disordered steady state within a single timestep, which we trace to a geometric symmetry argument that applies to any instantaneous, reflection-symmetric, position-based rule. Second, we replace hand-tuning of the hybrid with derivative-free optimization: CMA-ES searches the parameter space against a multi-metric scalar loss and finds a six-parameter hybrid (v4) that matches the three scalar metrics. Third, we introduce a near-zone anisotropy metric that the optimizer did not see, and use it to validate the calibrated models against the field's body-centered spatial structure. The near-zone metric separates v4 from a seven-parameter variant (v5) that has lower scalar loss but a forward-biased preferred-spacing ring; this out-of-sample test is what justifies treating v4 as the configuration that fits the field.

2 Background and related work

The Vicsek model [7] is a minimal self-propelled particle system: at each time step, every agent updates its heading to the arithmetic mean of neighbor headings inside an interaction radius, plus uniform angular noise of width η. Buhl et al. [2] fit this model to laboratory locust experiments and reported a phase transition from disorder to coherent marching as density increased. Bazazi et al. [1] proposed an alternative mechanism in which cannibalistic interactions drive forward motion: locusts in a marching band tend to bite one another, and individuals risk being bitten from behind, so contact and visual approach from behind elevate the propensity to march. Reduction of an individual's capacity to detect rear approach (via abdominal denervation) reduces its marching probability. Neither study had access to free-ranging field trajectories.

Weinburd et al. [8] computed body-centered neighbor density maps from their hopper-band data and observed an asymmetric distribution concentrated in the near zone (within roughly 3 cm of the focal), with more density behind the focal than ahead. They argued for directional interactions but stopped short of a specific mechanistic model.

Sayin et al. [6] fit a position-based model to field, laboratory, and virtual-reality experiments. In their pull formulation, each agent steers toward a weighted centroid of forward neighbors, with weights proportional to $\cos^2(\beta/2)$ for relative bearing β. They argued that bearing information is integrated over time by a neural ring-attractor circuit and that this temporal integration is necessary for the mechanism to produce coherent motion. Our v5 hybrid (Section 3) incorporates this temporal-integration component to test whether it is required to fit the Weinburd data.

The Couzin model [3] combines repulsion, alignment, and attraction in nested zones. We borrow the soft preferred-spacing idea from this literature for our hybrid extensions.

CMA-ES [4] is a derivative-free evolutionary optimizer that adapts a multivariate Gaussian sampling distribution as it explores. It is well suited to noisy black-box objectives in low-dimensional continuous spaces. We use the `pycma` implementation [5].

3 Methods

3.1 Field metrics

We compute four metrics from the Weinburd dataset and apply the same functions to every simulated trajectory:

1. Global polarization $\Phi(t) = \left|\frac{1}{N(t)}\sum_i e^{i\theta_i(t)}\right|$, averaged over time after burn-in.
2. Turning-angle standard deviation: per-step heading change $\Delta\theta_i(t) = \theta_i(t+1) - \theta_i(t)$, wrapped to $[-\pi, \pi]$, pooled across agents and time.
3. Median nearest-neighbor distance (NND).
4. Body-centered neighbor density. For each focal individual we rotate the relative position of every neighbor inside a 10 cm radius into the focal's body frame (heading along $+y$) and accumulate a 50×50 histogram. Figure 1 shows the field distribution.

From the density map we derive scalar anisotropy indices $A = (\rho_{\text{rear}} - \rho_{\text{front}})/(\rho_{\text{rear}} + \rho_{\text{front}})$, where ρ_{rear} and ρ_{front} are mean densities in the lower and upper halves of the body-centered histogram. By this convention $A > 0$ indicates more density behind the focal than ahead, consistent with the qualitative direction reported by Weinburd et al. [8]. We compute A at two scales: A_{global} averaged over the full half-disk out to 10 cm, and A_{near} restricted to a 1.5–3.0 cm radial band. The two-scale split matters because the anisotropy Weinburd et al. [8] report is concentrated in the near zone, so a near-isotropic value of A_{global} is compatible with non-trivial near-zone structure that the global metric does not resolve. Section 4.3 uses both indices: A_{global} as a coarse first check, A_{near} as the more discriminating spatial test. Crucially, the CMA-ES loss only sees the three scalar metrics; both anisotropy indices are computed post hoc, which makes A_{near} an out-of-sample validation of the calibrated models.

3.2 Models

The Vicsek model implements standard alignment dynamics with periodic boundaries. Each agent sets its next heading to $\theta_i(t{+}1) = \arg\sum_{j\in\mathcal{N}_i} e^{i\theta_j} + \eta\xi_i$, where $\mathcal{N}_i$ is the set of neighbors within $r_{\text{int}} = 7$ cm and $\xi_i \sim \mathcal{U}(-1/2, 1/2)$. Speed is constant and a hard repulsion override applies at $r_{\text{rep}} = 1.5$ cm.

The pull model has each agent steer toward the centroid of forward neighbors weighted by $w(\beta) = \cos^2(\beta/2)$, following Sayin et al. [6], but *without* the temporal-integration component. The noise term and the repulsion override match the Vicsek implementation. We test this simplified per-frame version explicitly to isolate the effect of temporal integration; v5 below restores it.

Hybrid v3 (anisotropic Vicsek) extends the alignment rule along two axes. Forward neighbors (relative bearing $|\beta| < \pi/3$) get weight $1 + \alpha_{\text{aniso}}$ and rear neighbors $1 - \alpha_{\text{aniso}}$. Effective noise scales with the fraction of neighbors ahead: $\eta_{\text{eff}}(i) = \eta_{\text{base}}(1 - \lambda f_{\text{forward}}(i))$. Speed varies with local order between $v_{\text{min}} = 2.7$ cm/s and $v_{\text{max}} = 11.76$ cm/s.

Hybrid v4 adds two mechanisms motivated by the v3 failure analysis (Section 4). The heading-inertia term has each agent commit a fraction $\mu \in (0, 1]$ of the way to its new desired heading θ^* via exponential smoothing in unit-vector space: $\theta_i(t{+}1) = \arg\left[(1{-}\mu)e^{i\theta_i(t)} + \mu e^{i\theta^*}\right]$. This narrows the per-step turning distribution. The preferred-spacing zone applies in the band $r_{\text{rep}} < d < r_{\text{pref}}$: the agent mixes a radial-push direction into its desired heading with weight k_{pref}.

Hybrid v5 adds temporal bearing integration after Sayin et al. [6]'s ring-attractor analogy. Each agent maintains an exponential moving average of the alignment-derived target unit vector, $T_{\text{ema}} \leftarrow$

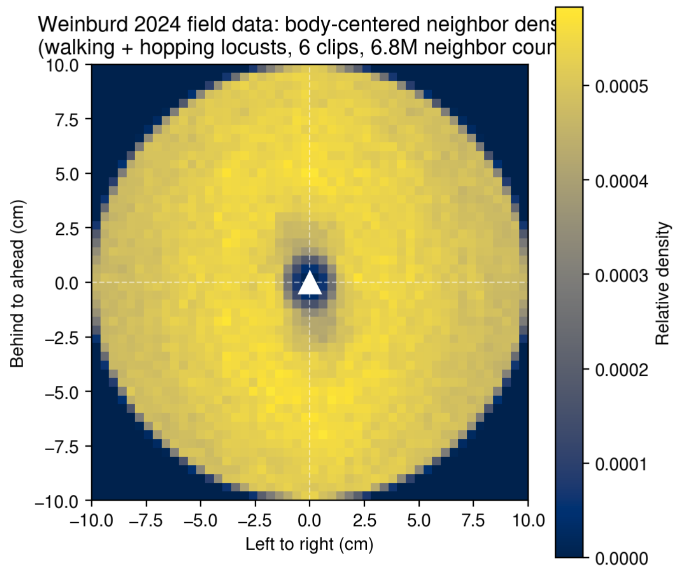

Figure 1: Body-centered neighbor density from the Weinburd et al. [8] hopper-band dataset, walking and hopping individuals combined (six clips, 6.86×10^6 neighbor counts). The focal heads up. The distribution is close to isotropic at the global level ($A_{\text{global}} = -0.0008$), with a self-exclusion hole at the origin and a near-zone anisotropy concentrated within ~ 3 cm of the focal ($A_{\text{near}} = -0.005$). We use this map and its scalar summaries as field-fitting targets alongside the three scalar metrics.

$(1-\rho)T_{\text{ema}} + \rho T_{\text{instantaneous}}$. The desired heading derives from T_{ema} rather than the instantaneous target. Setting $\rho = 1$ disables the integrator and recovers v4.

3.3 Calibration

We calibrate the hybrid models with CMA-ES against a weighted relative-error loss

$$L = w_\Phi \frac{|\Phi_{\text{sim}} - \Phi_{\text{field}}|}{\Phi_{\text{field}}} + w_\sigma \frac{|\sigma_{\text{sim}} - \sigma_{\text{field}}|}{\sigma_{\text{field}}} + w_{\text{NND}} \frac{|\text{NND}_{\text{sim}} - \text{NND}_{\text{field}}|}{\text{NND}_{\text{field}}}, \tag{1}$$

with weights $w_\Phi = 1$, $w_\sigma = w_{\text{NND}} = 2$. The two failure-mode metrics from the baseline experiments (Section 4) get double weight; polarization is already close in v3. This weighting is an experimenter choice. We test its sensitivity in Section 5 by re-running v4 with equal weights. Each evaluation runs five seeds of a 2000-step simulation with 500-step burn-in and returns the ensemble mean.

The optimization parameter space is six-dimensional for v4 and seven-dimensional for v5: $\eta_{\text{base}} \in [0.1, 1.5]$, $\lambda \in [0, 1]$, $\alpha_{\text{aniso}} \in [0, 1]$, $\mu \in [0.1, 1.0]$, $k_{\text{pref}} \in [0, 0.7]$, $r_{\text{pref}} \in [1.6, 5.0]$ cm, and (v5 only) $\rho_{\text{temporal}} \in [0.05, 1.0]$. Following the `pycma` guidance [5], we map every parameter linearly to the internal range $[0, 10]$ so a single σ_0 has uniform meaning across dimensions.

We run CMA-ES with population size 20, $\sigma_0 = 2.0$, and 40–50 generations. The 20 candidate evaluations within each generation are distributed across CPU cores using `multiprocessing.Pool`; BLAS threading inside workers is pinned to one to avoid oversubscription. On a 32-thread Intel i9-13900K the full v4 optimization completes in about 13 minutes (gen$_0$ in 19 seconds, end-to-end speedup of about $11\times$ over the serial implementation).

4 Results

4.1 The two baselines fail in distinct ways

After calibrating η to match field polarization, the Vicsek model hits $\Phi = 0.824$ (target 0.820) but produces a turning-angle distribution more than twice as wide as the field (0.595 rad vs 0.276 rad, 116% relative error) and undercounts NND by 33%. The model achieves order, but at the cost of pathological heading noise that compensates for the absence of any heading-stabilizing mechanism beyond raw alignment.

The pull model fails differently, and more sharply. In every parameter combination we tested – a pull-strength sweep across $\{0.3, 0.5, 0.7, 0.8, 0.9, 1.0\}$ at fixed noise, and a 29-point noise sweep from 0.1 to 2.9 rad at pull strength 1.0, with cone half-angle held at $\pi/2$ – polarization remained at $\Phi \approx 0.04$–0.06, with a turning-angle distribution near 2 rad (essentially saturated). The failure is not a tuning failure: it persists across the entire parameter sweep we tested.

The failure is also not a bootstrapping failure. Initialized from perfect order ($\Phi_0 = 1.000$, headings drawn with std 0.01 rad across five seeds), polarization drops 96% within a single timestep and converges to the same disordered steady state as random initialization (mean 0.0372 vs 0.0367, both with seed std $\lesssim 10^{-3}$). The model destroys existing order; it does not merely fail to generate it.

The cause is geometric. In a disordered swarm, neighbor bearings are approximately uniformly distributed around the focal individual. The forward-weighting function $\cos^2(\beta/2)$ is symmetric under the reflection $\beta \to -\beta$. Any rule whose weighting is symmetric under that reflection produces, in expectation over an isotropic bearing distribution, zero net lateral steering signal. The forward component is non-zero, so each agent maintains its current heading; but there is no mechanism to bias the population toward a common direction. Small stochastic deviations from isotropy provide a finite-N signal in principle, but heading noise randomizes each agent's direction faster than these deviations can accumulate into coherent motion. The result is a steady state at $\Phi \approx 0.037$ that closely matches the finite-N baseline expected for $N = 150$ agents moving independently with no effective coupling.

This argument rules out an entire class of instantaneous, reflection-symmetric, position-based steering rules as candidate mechanisms for sustaining order in a disordered swarm. It does not rule out the full Sayin et al. [6] proposal, which adds temporal integration via a ring-attractor circuit. The argument is in fact consistent with their emphasis on the ring attractor: temporal integration is what their formulation requires precisely because the per-frame steering rule on its own cannot do the work. Hybrid v5 (Section 4.2) tests whether adding temporal integration to a hybrid alignment-plus-pull model improves the fit to the Weinburd data.

Hybrid v3 (anisotropic Vicsek with hand-tuned $\eta = 1.1$, $\lambda = 0.8$, $\alpha = 0.6$) hits polarization within 0.5% but still misses turning-angle std by 88% and NND by 45%. Forward weighting tightens the alignment, which lowers η enough to bring polarization in at a sensible noise level, but introduces no mechanism that limits per-step heading change or sets a preferred spacing.

4.2 CMA-ES closes the scalar gaps

Figure 2 shows the loss trajectories. The v4 run reaches a final loss of 0.099. The calibrated ensemble means are $\Phi = 0.898$ (9.6% above the field target), $\sigma = 0.276$ rad, and NND median $= 3.90$ cm. The two failure modes that motivated heading inertia and preferred spacing close to within seed-to-seed variation: the across-seed std on σ is 0.010 rad for v4 and 0.004 rad for v5 (roughly 3.6% and 1.4% of the field value), and on NND is 0.049 cm for v4 and 0.045 cm for v5 (roughly 1.3% in both cases). We deliberately do not report percentage errors smaller than seed noise. Polarization remains slightly elevated; the 9.6% residual is a direct consequence of our weighting choice (Section 5).

The v5 run reaches a final loss of 0.085, but the converged $\rho_{\text{temporal}} = 0.95$ is near the no-effect end

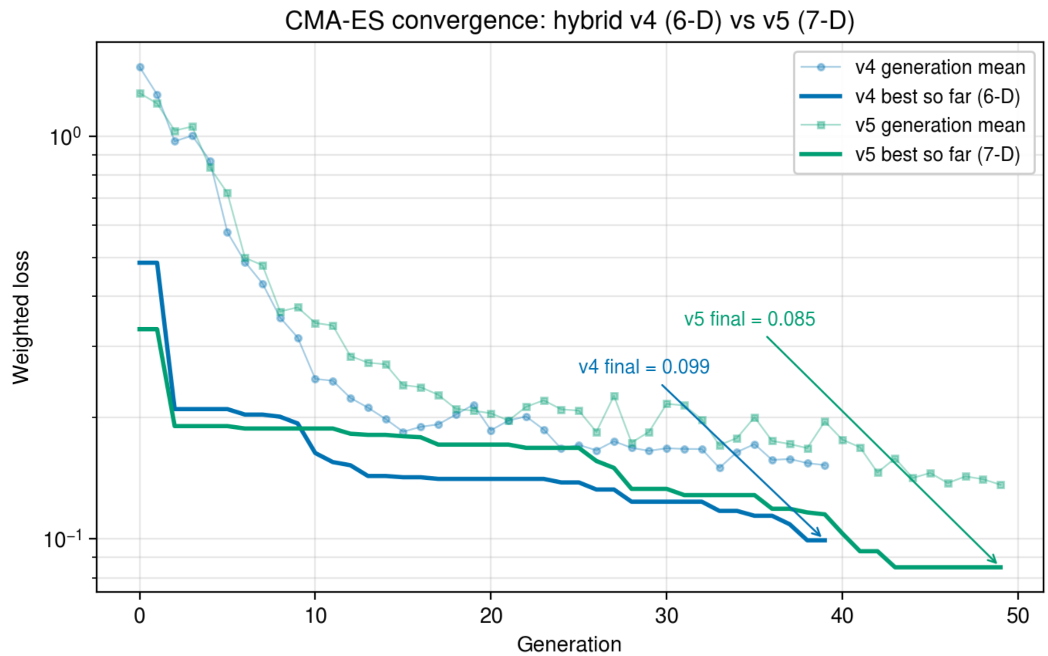

Figure 2: CMA-ES convergence curves. Both v4 (six-dimensional) and v5 (seven-dimensional) reduce the weighted loss by an order of magnitude over 40–50 generations. The v5 run plateaus around generation 25 before a late breakout finds a slightly better basin. As Section 4.3 shows, the basin v5 escaped into is worse on the out-of-sample spatial metric.

of its allowed range (recall $\rho = 1$ disables the integrator). The lower scalar loss does not come from the integration mechanism. It comes from the optimizer finding a different basin: α_{aniso} saturates at 1.0 and λ drops to 0.17, while the other parameters move only slightly. The two CMA-ES-calibrated runs separate further once the spatial structure is examined directly (Section 4.3): v4 reproduces the field's near-zone density, v5 does not.

Figure 3 summarizes per-metric performance for every model we ran. Numerical values are listed in Table 1.

4.3 Spatial structure: v4 matches, v5 distorts

The body-centered neighbor density map is the test that motivated the hybrid extensions in the first place. Weinburd et al. [8] reported anisotropic neighbor distributions in their data, and our hybrid v3 appeared to reproduce a visible asymmetry in the inner ring of its density map.

The global anisotropy index is essentially zero in the field ($A_{\text{global}} = -0.0008$). v3, v4, and v5 all reproduce the global value to within $|A| < 0.001$ (Figure 4, panel titles). Match on A_{global} alone is therefore a weak test: all the calibrated models pass, but they pass by all being near-isotropic at the global scale rather than by reproducing structured signal.

The near-zone metric A_{near}, restricted to a 1.5–3.0 cm radial band, separates the calibrated hybrids. The field value is -0.005, marginally favoring rear-side density. v3 returns -0.003 and v4 returns -0.005, both close to the field. v5's value is -0.061, an order of magnitude larger, with substantially more density ahead of the focal than behind. The preferred-spacing ring at v5's optimum sits closer to the focal in front than at the rear, at radii where the field shows no asymmetry. The white dotted circles in Figure 4 mark the band.

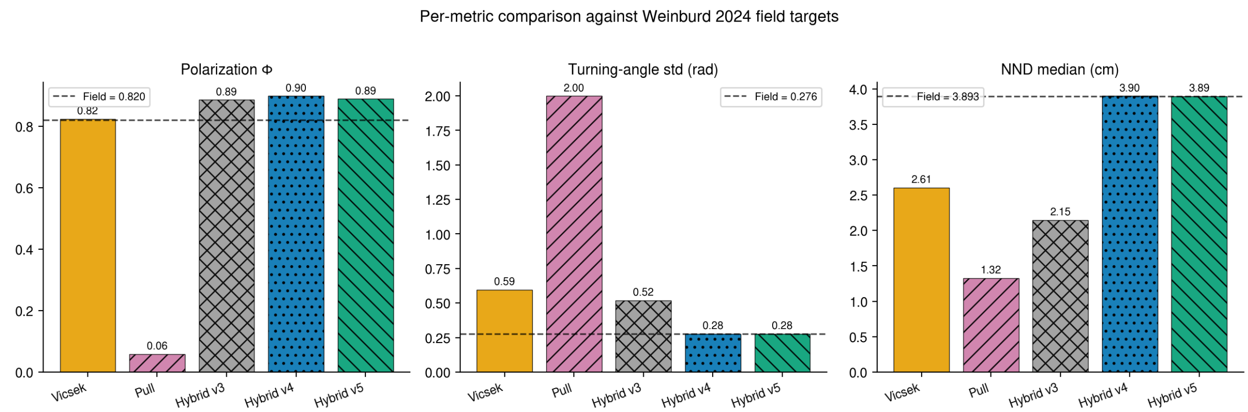

Figure 3: Field-fit scorecard across all models on the three scalar metrics. Dashed lines indicate field targets. Vicsek and Pull miss multiple metrics. Hybrid v3 closes polarization but not the others. CMA-ES-calibrated v4 and v5 hit all three scalar targets within seed noise (under periodic boundary conditions, see Section 4.4). The body-centered spatial structure separates v4 from v5; see Figure 4.

Table 1: Field-fit scorecard. Relative error in parentheses. Field targets: $\Phi = 0.820$, $\sigma = 0.276$ rad, NND $= 3.893$ cm. $\pm$ values on Φ are the across-seed standard deviation of the per-seed mean (10 seeds for the baseline runs, 5 seeds for the CMA-ES-calibrated runs). Errors on σ and NND for the calibrated hybrids fall below the seed-to-seed standard deviation; we report them as $<$1% for completeness but do not interpret sub-seed-noise residuals.

Model	Φ $\pm\sigma_{\text{seed}}$	σ (rad)	NND (cm)
Vicsek	0.824 ± 0.019 (0.5%)	0.595 (116%)	2.61 (33%)
Pull	0.058 ± 0.001 (93%)	1.999 (624%)	1.32 (66%)
Hybrid v3	0.890 ± 0.025 (8.5%)	0.520 (88%)	2.15 (45%)
Hybrid v4	0.898 ± 0.027 (9.6%)	0.276 ($<$1%)	3.90 ($<$1%)
Hybrid v5	0.889 ± 0.012 (8.4%)	0.276 ($<$1%)	3.89 ($<$1%)

The CMA-ES loss did not include A_{near}, so the optimizer could place near-zone structure anywhere that did not break the three scalar metrics. v4 found a basin in which the preferred-spacing ring sits symmetrically around the focal ($\alpha_{\text{aniso}} = 0.80$, $\lambda = 0.52$). v5, with the extra ρ_{temporal} dimension, found a different basin: α_{aniso} saturates at 0.99 and λ drops to 0.17, which sharpens the ring and biases it forward. v5's scalar loss is 0.085 versus v4's 0.099, so the optimizer prefers v5; on the out-of-sample spatial metric the trade goes the other way. The seventh parameter is unnecessary at the field's measurement scale, as Section 4.2 already noted, and in our calibration it actively worsens the spatial fit. v4 is the configuration that matches all four metrics; v5 is a worse fit dressed up as a slightly better one.

The "forward void" claim attached to earlier v3 figures was an overinterpretation of a near-zone feature that is small in the field ($A_{\text{near}} = -0.005$). v3 reproduces the field's near-zone structure for the same reason it fails the scalar metrics: forward-weighted alignment alone is enough for spatial structure but not for tight heading control or NND.

4.4 Boundary-condition dependence

We tested the calibrated v4 model under reflective walls instead of periodic boundaries, holding all other parameters fixed. The fit broke: turning-angle std jumped to 0.493 rad (78% off) and NND dropped to

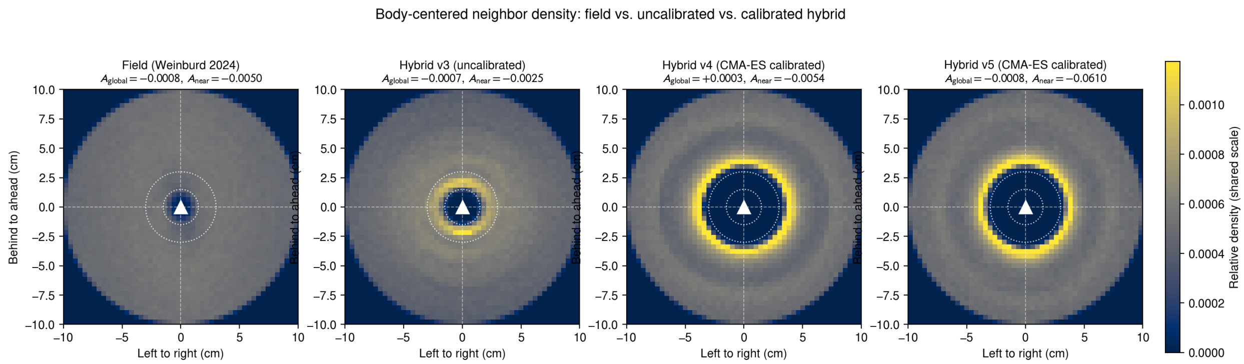

Figure 4: Body-centered neighbor density for the field and the three hybrid models, on a shared color scale. The global anisotropy index is within $|A| < 0.001$ of zero in all four. The near-zone index, computed only inside the white dotted band ([1.5, 3.0] cm), separates the v5 ring artifact ($A_{\text{near}} = -0.061$) from the near-flat field (-0.005), v3 (-0.003), and v4 (-0.005). v4 fits both scalar and near-zone metrics; v5 trades the spatial fit for a marginally lower scalar loss.

3.47 cm (11% off). Polarization moved closer to target, from 0.898 to 0.752 (8.3% off). The periodic image structure is doing real physical work in the calibration. Reflective walls perturb headings on contact, inflating the per-step turning distribution, and they compress densities away from the boundary, lowering NND.

This is a substantive limitation, not a footnote. Real hopper bands are open systems with front and rear asymmetry, not periodic boxes. The calibrated parameters in Table 1 are conditional on a boundary regime that does not match the system we are modeling. The same hybrid mechanism, calibrated under different boundary conditions, would converge on different parameters and almost certainly different fits. We have demonstrated that with six free parameters, CMA-ES can match the Weinburd scalar metrics and the near-zone anisotropy under periodic boundaries; we have *not* demonstrated that this rule structure governs the dynamics of free-ranging locust hopper bands. Establishing that would require calibration under realistic open-boundary conditions, which we did not have time to implement. We discuss possible approaches in Section 5.

5 Discussion

The pull-model failure result and its geometric explanation stand independently of the calibration story. They establish that no instantaneous, reflection-symmetric, position-based steering rule can sustain order in a disordered swarm, regardless of cone angle, pull strength, or noise level, and they point directly at temporal integration (or some other symmetry-breaking mechanism) as the missing ingredient. This is consistent with Sayin et al. [6]'s mechanistic emphasis on the ring-attractor circuit, and it tightens the claim: the ring attractor is not just one possible implementation of pull dynamics but a computationally necessary component of any pull-class mechanism that generates order.

We spent several weeks hand-tuning a hybrid model and producing scorecards that captured one metric well at a time. CMA-ES, with parallel evaluation across cores, runs end-to-end in about 13 minutes and finds parameter combinations that hand-tuning missed. For noisy black-box objectives in this dimensionality, derivative-free evolutionary search is the right default. We should have used it sooner.

Adding temporal bearing integration (ρ_{temporal}) gave the optimizer a seventh degree of freedom that it drove near its no-effect limit ($\rho = 0.95$). On the scalar loss alone the change looks benign and even slightly helpful, but the extra dimension shifted the converged basin: α_{aniso} saturated at 1.0 in v5 against v4's 0.80, and that re-shaped the preferred-spacing ring into the forward-biased configuration that breaks the near-zone fit (Section 4.3). Its geometrically established necessity (Section 4) for any *pure* pull-class model remains. What our v5 result shows is narrower: at the granularity of the Weinburd et al. [8] dataset, and within an alignment-dominated hybrid that already breaks reflection symmetry through forward-weighted coupling, the additional temporal-integration term is unnecessary, and in our calibration harmful. A measurement that distinguishes instantaneous from time-integrated bearing estimation would be needed to test the ring-attractor mechanism directly; neither our scalar nor our spatial metrics do.

The remaining gap is polarization. The v4 and v5 calibrations sit at $\Phi \approx 0.89$ versus a field target of 0.82. We re-ran v4 with equal weights ($w_\Phi = w_\sigma = w_{\text{NND}} = 1$) to test whether the gap was an artifact of our prioritized weighting. The equal-weight optimum lands at $\Phi = 0.883$, $\sigma = 0.277$ rad, NND $= 3.92$ cm, with λ dropping from 0.52 to 0.24 and α_{aniso} saturating at 0.99. The polarization gap narrows by 0.015 and the other two metrics do not break. v4's polarization floor is therefore close to 0.88 regardless of weighting, which we read as a structural property of the model at this density rather than a tunable parameter.

The v3 anisotropy reframing in Section 4.3 is the most significant correction relative to our earlier work on this problem. The visible v3 asymmetry that motivated the hybrid extensions was real but near-zone; our initial A_{global} index did not capture it; and matching on A_{global} alone is matching at zero rather than matching on a structured signal. Introducing A_{near} as an out-of-sample test changes the picture: v4 passes it, v5 fails it, and the calibrated v4 becomes a model that matches all four metrics rather than three.

We have not run an open-boundary calibration. The reflective-wall stress test indicates the trade-offs differ. Three plausible follow-ups are: a flux-boundary scheme that despawns leaving agents and respawns at the rear; calibration under reflective walls with the loss reweighted; or a sliding-window simulation centered on the swarm centroid. Of these, we think the sliding-window approach is most defensible because it reproduces the actual measurement geometry of the Weinburd et al. [8] field recordings, but we did not have time to implement any.

6 Project contributions

Asher Albrecht built the field-data ingestion pipeline from the Weinburd et al. [8] `.mat` files, computed the four baseline metrics on the field data (Section 3.1), implemented the Vicsek and Pull baseline models, ran the pull-model order-destruction experiment and geometric analysis (Section 4), and led the hybrid v3 anisotropic-Vicsek design and the η/λ parameter sweep figures.

George Stephenson extended the hybrid model with heading inertia, preferred spacing, and temporal bearing integration (v4 and v5 in `hybrid_v4.py` and `hybrid_v5.py`), built the parallel CMA-ES calibration framework (`calibrate_cma.py`, `calibrate_cma_v5.py`, `validate_calibrated.py`), implemented the global and near-zone spatial-anisotropy validation against the field density map, and produced the post-calibration figures.

Both authors contributed to the experimental design, the slide deck, and this paper.

Code and data

The simulation code, calibration scripts, CMA-ES log files, and figure generators are at `https://github.com/gsstephenson/locust-collective-motion`. The Weinburd dataset is available from the original authors at Weinburd et al. [8].

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